

Analysis Data Reviewer's Guide

Sponsor Name; Study CDISCPILLOT01; ADRG Template Version ccyy-mm-dd

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1 Introduction

1.1 Purpose

This document provides context for the analysis datasets and terminology that benefit from additional explanation beyond the Data Definition document (define.xml) for an individual study. In addition, this document provides a summary of ADaM conformance findings.

1.2 Acronyms

Acronym	Translation
aCRF	Annotated Case Report Form
ADaM	Analysis Dataset Model
ADRG	Analysis Data Reviewer's Guide
eCRF	Electronic Case Report Form
eDT	Electronic Data Transfer (e.g. central lab data, ECG vendor data, PK data, etc.)
IG	Implementation Guide
NA	Not Applicable
SDTM	Study Data Tabulation Model
TAUG	Therapeutic Area User Guide

2 Dataset Standards

Standard or Dictionary	Versions Used
SDTM	SDTM Implementation Guide Version 3.1.2 ; SDTM Version 1.2
Medical Events Dictionary	MedDRA version 8.0
Define-XML	Define version 1.0.0

2.1 Source Data Used for Analysis Dataset Creation

{source data description}

Include the following text if applicable: Please refer to the Legacy Data Conversion Plan and Report Appendix for additional details.

3 Protocol Description

3.1 Protocol Number and Title

Protocol Number: CDISCPLOT01

Protocol Title: Safety and Efficacy of the Xanomeline Transdermal Therapeutic System (TTS) in Patients with Mild to Moderate Alzheimer’s Disease

Protocol Versions:

[{‘description’: ‘Protocol with 3 amendments (numbered not provided in report).’, ‘details’: ‘The protocol was amended three times. The first two amendments changed ambulatory ECG assessments (shortened the 48-hour monitoring to 24 hours prior to Visit 3, added monitoring prior to Visit 4, later removed monitoring prior to Visit 6, and subsequently removed the monitoring prior to Visit 4). While additional ECG monitoring prior to Visit 4 was required, an extra visit (designated Visit 3e) was added for placement of ambulatory ECG equipment.’, ‘effect_on_data_analysis’: ‘Changes to protocol-specified analyses were made and are described in the Statistical Analysis Plan (Appendix 9). These included omission of some secondary/sensitivity analyses and some efficacy and safety endpoints and the inclusion of additional types of analyses. (No amendment/version numbers or dates are provided in the report.)’}]

3.2 Protocol Design in Relation to ADaM Concepts

3.2.1 Protocol Objective

{‘primary_objectives’: [‘Determine if there is a statistically significant relationship (overall Type I error rate = 0.05) between change in both the ADAS-Cog (11) and CIBIC+ scores and drug dose (0, 50 cm2 [54 mg], and 75 cm2 [81 mg]).’, ‘Document the safety profile of the xanomeline transdermal therapeutic system (TTS).’], ‘secondary_objective’: [‘Assess dose-dependent improvement in behavior as measured by the Revised Neuropsychiatric Inventory (NPI-X; mean from Week 4 to Week 24).’]}

3.2.2 Protocol Methodology

{‘study_design_type’: ‘Prospective, randomized, multi-center, double-blind, placebo-controlled, parallel-group study (Phase 2).’, ‘randomization’: ‘Subjects were randomized equally (1:1:1) to three groups: placebo, xanomeline low dose, xanomeline high dose.’, ‘blinding’: ‘Double-blind (investigators and subjects blinded). Placebo patches matched active patches in appearance.’, ‘treatment_and_dosing’: ‘Two transdermal patches applied daily (one 50 cm2 patch = 54 mg; one 25 cm2 patch = 27 mg). Treatment arms: placebo (both placebo patches = 0 mg),

xanomeline low dose (50 cm2 active + 25 cm2 placebo = 54 mg total), xanomeline high dose (both active patches = 81 mg total).’, ‘treatment_duration_and_visits’: ‘26 weeks on-study with 24 weeks of active treatment; scheduled visits at screening (Week -2) and Weeks 0, 2, 4, 6, 8, 12, 16, 20, 24 and follow-up at Week 26 (visit numbering per protocol; an extra visit 3e was added during a period of extra ECG monitoring).’, ‘primary_endpoints_and_assessments’: ‘Co-primary efficacy endpoints: ADAS-Cog (11) change at Week 24 and CIBIC+ at Week 24. Secondary: ADAS-Cog (11) and CIBIC+ at Weeks 8 & 16; mean NPI-X from Week 4 to Week 24. Safety endpoints: adverse events, vital signs (weight, standing and supine blood pressure, heart rate), laboratory evaluations.’, ‘statistical_methodology’: “Hypothesis testing at significance level 0.05. Primary efficacy analyses used the efficacy population with LOCF imputation for missing Week 24 values. ADAS-Cog (11) primary analysis: ANCOVA (baseline score, site, treatment) with treatment coded as continuous for dose-response test; supportive analysis: likelihood-based repeated measures (MMRM). CIBIC+: ANOVA (site and treatment) with treatment as continuous for dose-response. NPI-X: ANCOVA (baseline, site, treatment) using mean of Weeks 4–24. Safety analyses: AE summaries by SOC/PT (MedDRA coding masked), Fisher’s exact test for comparisons versus placebo, Kaplan–Meier for time to first dermatologic event, CMH tests stratified by baseline status for lab shift analyses. Populations defined: ITT (all randomized), Safety (randomized and dosed), Efficacy (safety population with at least one post-baseline ADAS-Cog and CIBIC+), Completers (efficacy subjects completing Week 24).”, ‘imputation_and_handling_of_missing_data’: ‘Primary analyses used LOCF for missing Week 24 values; MMRM used as supportive repeated-measures approach; other deviations from protocol-specified analyses are documented in the SAP (Appendix 9).’}

3.2.3 Number of Subjects Planned in Total and by Group

{‘planned’: {‘total’: 300, ‘breakdown_by_group’: {‘placebo’: 100, ‘xanomeline_low_dose’: 100, ‘xanomeline_high_dose’: 100}}, ‘actual_enrolled_randomized’: {‘total_randomized’: 254, ‘breakdown_by_group’: {‘placebo’: 86, ‘xanomeline_low_dose’: 84, ‘xanomeline_high_dose’: 84}}}

3.2.4 Study Design Schema

{‘summary’: ‘Parallel-group randomized controlled trial with three treatment arms (placebo, xanomeline low dose, xanomeline high dose). Screening at Week -2, randomization at Week 0, active treatment through Week 24, study completion/visit at Week 26. Subjects applied two transdermal patches daily for the treatment period.’, ‘treatment_groups’: [{‘name’: ‘Placebo’, ‘description’: ‘Both 50 cm2 and 25 cm2 patches placebo (0 mg).’}, {‘name’: ‘Xanomeline low dose’, ‘description’: ‘50 cm2 active (54 mg) + 25 cm2 placebo (0 mg) = 54 mg total.’}, {‘name’: ‘Xanomeline high dose’, ‘description’: ‘50 cm2 active (54 mg) + 25 cm2 active (27 mg) = 81 mg total.’}], ‘phases_and_periods’: {‘screening’: ‘Week -2 (Visit 1).’, ‘randomization’: ‘Week 0 (Visit 3 in protocol numbering); first patch application at randomization.’, ‘treatment_period

(on-treatment / analysis period): ‘Post-randomization through Week 24 (planned primary analysis window). For lab analyses the treatment period is defined as any planned visit after Week 0 up to and including Week 24.’, ‘follow_up’: ‘Week 26 (Visit 13) follow-up/completion visit.’}, ‘analysis_populations_and_adam_mapping’: {‘treatment_variable_concept’: “Randomized treatment assignment (placebo / xanomeline low / xanomeline high) used as treatment in analyses (ADaM ‘treatment’ variable equivalent).”, ‘analysis_periods’: [‘On-treatment analysis period: post-randomization through Week 24 (primary efficacy/safety analyses).’, ‘Post-treatment/follow-up period: Week 24 to Week 26 (safety follow-up window).’], ‘analysis_phases_and_methods’: ‘Primary efficacy analyses used the Efficacy population with LOCF to Week 24; supportive MMRM repeated-measures analyses used post-baseline data across time points (visits) to model treatment-by-time effects. NPI-X endpoint computed as the mean of available scores Weeks 4–24 (analysis uses that treatment period). Safety datasets/analyses used Safety population (all randomized and dosed).’, ‘populations’: {‘ITT’: ‘All randomized subjects (used for disposition summaries).’, ‘Safety’: ‘All randomized subjects known to have taken at least one dose (used for safety analyses).’, ‘Efficacy’: ‘Subjects in safety population with at least one post-baseline ADAS-Cog and CIBIC+ (used for primary efficacy analyses).’, ‘Completers’: ‘Efficacy subjects who completed Week 24 (used for completers analyses).’}}, ‘visit_schedule_summary’: ‘Visits at Screening (Week -2), randomization/first dose at Week 0, scheduled assessments at Weeks 2, 4, 6, 8, 12, 16, 20, 24 (end of treatment), and Week 26 (end of study). An extra Visit 3e was added transiently for ambulatory ECG placement when additional ECG monitoring was required by amendments.’}

4 Analysis Considerations Related to Multiple Analysis Datasets

4.1 Core Variables

Core variables are those that are represented across all/most analysis datasets.

Variable Name	Variable Description
AGE	Age
AGEGR1	Pooled Age Group 1
AVAL	Analysis Value
AVISITN	Analysis Visit (N)
BASE	Baseline Value
BMIBL	Baseline BMI (kg/m ²)
CHG	Change from Baseline
CNSR	Censor
HEIGHTBL	Baseline Height (cm)
MMSETOT	MMSE Total
PARAM	Parameter
PARAMCD	Parameter Code

Variable Name	Variable Description
RACE	Race
STUDYID	Study Identifier
TRT01A	Actual Treatment for Period 01
TRTPN	Planned Treatment (N)
USUBJID	Unique Subject Identifier
WEIGHTBL	Baseline Weight (kg)

4.2 Treatment Variables

- ARM versus TRTxxP: Are the values of ARM equivalent in meaning to values of TRTxxP? **Yes.** Evidence: the ADaM mapping text (“Randomized treatment assignment ... used as treatment in analyses (ADaM ‘treatment’ variable equivalent)”) and the analysis code that groups by TRTP and sets the population treatment variable to TRT01P. TRTPN values (e.g., 0, 81) are used in filters, confirming TRTxxP encodes the randomized/planned treatment.
- ACTARM versus TRTxxA: If TRTxxA is used, then are the values of ACTARM equivalent in meaning to values of TRTxxA?
- Use of ADaM Treatment Variables in Analysis: Are both planned and actual treatment variables used in analysis? **Yes.** Planned treatment variables are used for efficacy tables/summary grouping and population assignment; actual treatment is used in at least the safety/time-to-event (KM) analysis per the supplied scripts.
- Use of ADaMTreatment Grouping Variables in Analysis: Are both planned and actual treatment grouping variables used in analysis? **Yes.** Note: there are minor naming inconsistencies in the sample code (e.g., `set_pop_treat_var(TRT01P)` vs `TRT01A`), but the provided programs and dataset variable lists demonstrate that planned (TRTP/TRTPN) and actual (TRT01A) treatment grouping variables are both employed in analyses.

4.3 Subject Issues that Require Special Analysis Rules

(insert your text here or indicate that there were no subject issues to be documented)

4.4 Use of Visit Windowing, Unscheduled Visits, and Record Selection

- Was windowing used in one or more analysis datasets? **Yes.** Windowing is therefore used (at least) in the ADADAS analysis dataset to select the single visit record closest to the target date when multiple visits fall in the same window.

- Were unscheduled visits used in any analyses? **Yes.** This means unscheduled or extra visits were eligible and would be used when they were the record closest to the nominal target date (and thus flagged ANL01FL='Y'). I cannot state how many or which subjects' records came from unscheduled visits without the ADADAS/ADLB source data showing ANL01FL and visit origin.
- Additional Content of Interest <See ADRG Completion Guidelines for additional content of interest, and include text here or remove this text >.

4.5 Imputation/Derivation Methods

- If date imputation was performed, were there rules that were used in multiple analysis datasets?
- Additional Content of Interest <See ADRG Completion Guidelines for additional content of interest, and include text here or remove this text >.

5 Analysis Data Creation and Processing Issues

5.1 Split Datasets

{split datasets description}

5.2 Data Dependencies

dataset name	depend on the following datasets
ADADAS	
ADAE	ADSL
ADLBC	ADSL
ADSL	
ADTTE	ADAE, ADSL

5.3 Intermediate Datasets

{intermediate datasets description}

6 Analysis Dataset Descriptions

6.1 Overview

- Are data for screen failures, including data for run-in screening (for example, SDTM values of ARMCD='SCRNFAIL', or 'NOTASSGN') included in ADaM datasets? **No.** Screen-failure records remain in SDTM (e.g., DM/DS) per the define metadata; if screen-failure-level ADaM outputs are required they would need to be created separately (not present in the supplied ADaM spec).
- Are data taken from an ongoing study?
- Do the analysis datasets support all protocol-and statistical analysis plan-specified objectives?
- Include all objectives listed in the protocol or SAP which are not supported in the analysis datasets and the reason for their absence.

Additional Content of Interest

(See ADRG Completion Guidelines for additional content of interest, and includetext here or remove this text).

6.2 Analysis Datasets

Dataset

Dataset Label	Class	Efficacy	Safety	Baseline or other subject characteristics	PK/PD	Primary Objective	Structure
ADAS-Cog Analysis ADADAS	BASIC DATA STRUC-TURE	X					one record per subject per parameter per analysis visit per analysis date

Dataset Label	Class	Efficacy	Safety	Baseline or other subject characteristics	PK/PD	Primary Objective	Structure
Adverse Events Analysis Dataset ADAE	ADAM OTHER		X				one record per subject per adverse event
Analysis Dataset Lab Blood Chemistry ADLBC	BASIC DATA STRUCTURE	X	X				one record per subject per parameter per analysis visit
Subject-Level Analysis Dataset ADSL	SUBJECT LEVEL ANALYSIS DATASET			X			one record per subject. Screen Failures are excluded.
AE Time To 1st Derm. Event Analysis ADTTE	BASIC DATA STRUCTURE		X				one record per subject per parameter

6.2.1 ADSL – Subject Level Analysis Dataset

{adsl dataset description}

{date imputation rules}

6.2.2 Dataset – Dataset Label

(A new section is required for each dataset that is hyperlinked in the inventory table. This section should be copied to create a new section for each dataset. The text in the section header above must be edited to match the dataset name and label.

7 Data Conformance Summary

7.1 Conformance Inputs

Specify the software name and version for the analysis datasets

(Text here)

Specify the version of the validation rules (i.e. CDISC, FDA) for the analysis datasets

(Text here)

Specify the software name and version for the define.xml

(Text here)

Specify the version of the validation rules (i.e. CDISC, FDA) for the define.xml

(Text here)

Provide any additional compliance evaluation information:

(Text here)

7.2 Issues Summary

(insert your text here and/or use following table)

Dataset	Diagnostic Message	Severity	Count	Explanation
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8 Submission of Programs

All programs for analysis datasets and primary and secondary efficacy results are submitted. They were all created on a platform using. The internal reference date used to create dates in ADaM datasets is.

8.1 ADaM Programs

Program Name	Output	Dataset Description
adadas	adadas.rds	ADAS-Cog Analysis
adae	adae.rds	Adverse Events Analysis Dataset
adlbc	adlbc.rds	Analysis Dataset Lab Blood Chemistry
adsl	adsl.rds	Subject-Level Analysis Dataset
adtte	adtte.rds	AE Time To 1st Derm. Event Analysis

8.2 Analysis Output Programs

r_file	outputs	filters	variables
tlf-demographic.r	tlf-demographic-pilot5.out	ADSL.STUDYID == 'CDISCPIL0T01'; ADSL.ITTFL == 'Y'	ADSL.AGE; ADSL.AGEGR1; ADSL.RACE; ADSL.HEIGHTBL; ADSL.WEIGHTBL; ADSL.BMIBL; ADSL.MMSETOT
tlf-efficacy.r	tlf-efficacy-pilot5.rtf	ADLB.TRTPN IN (0, 81); ADLB.PARAMCD == 'GLUC'; ADLB.AVISITN IS NOT NULL; ADLB.AVISITN == 20; ADLB.AVISITN == 0; ADLB.AVISITN == 20 AND ADLB.CHG IS NOT NULL AND ADLB.BASE IS NOT NULL	ADSL.STUDYID; ADSL.USUBJID; ADLB.BASE; ADLB.TRTPN; ADLB.PARAMCD; ADLB.AVISITN; ADLB.CHG; ADLB.AVAL

r_file	outputs	filters	variables
tlf-kmplot.r	pdf/tlf-kmplot-pilot5.pdf	ADSL.SAFFL == 'Y'; ADSL.STUDYID == 'CDISCPILLOT01'; ADTEE.PARAMCD == 'TTDE'; ADTEE.STUDYID == 'CDISCPILLOT01'	ADSL.STUDYID; ADSL.USUBJID; ADSL.TRT01A; ADTE.PARAMCD; ADTE.AVAL; ADTE.CNSR; ADTE.PARAM
tlf-primary.r	tlf-primary-pilot5.rtf	ADAS.EFFFL == 'Y'; ADAS.ITTFL == 'Y'; ADAS.PARAMCD == 'ACTOT'; ADAS.ANL01FL == 'Y'; ADSL.EFFFL == 'Y'; ADSL.ITTFL == 'Y'	ADAS.AVAL; ADAS.CHG

8.3 Open source R packages

Package	Version	Description
admiral	1.3.0	This R package provides tools for creating CDISC-compliant Analysis Data Model (ADaM) datasets, essential for regulatory submissions to the FDA, by implementing required analysis derivations as per the ADaM Implementation Guide.
admiraldev	1.3.1	This package provides utility functions for checking data, variables, and conditions in 'admiral' and its extension packages, along with additional tools to aid developers in documentation, testing, and maintenance.

Package	Version	Description
assertthat	0.2.1	This package enhances the typical <code>stopifnot()</code> function by allowing users to easily specify preconditions and postconditions for their code, while also generating clear and informative error messages for easier debugging.
backports	1.5.0	This package provides re-implemented functions and features from R v3.0.0 and later, enabling compatibility with older installations by conditionally exporting backports.
base64enc	0.1-3	This package provides flexible tools for encoding and decoding data in base64 format, enhancing functionality beyond that of the discontinued ‘base64’ package.
bit	4.6.0	This package offers efficient classes and methods for boolean and skewed boolean vectors, rapid integer sorting, set operations, and foundational tools for handling large datasets with features like range indexing and chunked processing.
bit64	4.6.0-1	The ‘bit64’ package in R provides support for serialized 64-bit signed integers, enabling precise handling of large numeric values and database keys, while offering various methods for coercion and integration with standard data structures.

Package	Version	Description
brew	1.0-10	This package provides a templating framework that combines text and R code for generating reports, using a syntax similar to templates in PHP, Ruby, Java, and Python.
broom	1.0.8	The Broom package in R provides tools to neatly summarize and extract information from statistical models, allowing users to easily report results, create visualizations, and manage multiple models effectively.
cachem	1.1.0	This package provides key-value stores with automatic pruning capabilities, allowing caches to enforce limits on total size or object age by automatically removing older or excessive entries.
callr	3.7.6	This package allows users to execute computations in an isolated R process, ensuring that the current R session remains unaffected.
cellranger	1.1.0	This package provides helper functions for manipulating spreadsheets, focusing on the “A1:D10” format for specifying cell ranges.
checkmate	2.3.2	This package provides efficient tests and assertions for frequent argument checks, utilizing C to minimize execution time overhead.

Package	Version	Description
cli	3.6.5	This package provides tools for creating visually appealing command line interfaces (CLIs) using semantic elements and customizable themes, along with support for ANSI colors and various text styles.
clipr	0.8.0	This package provides simple utility functions for reading from and writing to the clipboard across Windows, OS X, and X11 environments.
commonmark	2.0.0	This package utilizes the ‘cmark’ reference implementation to convert markdown text into various formats such as HTML, LaTeX, and groff man, while also providing access to the markdown parse tree in XML format and supporting GFM extensions like tables and strikethrough text.
cowplot	1.2.0	This R package enhances ‘ggplot2’ by providing tools for creating publication-quality figures, including themes, alignment functions, and annotation features.
cpp11	0.5.2	The ‘cpp11’ package offers a safe, modern C++11 interface to R’s C API, ensuring compliance with R function semantics, support for ‘ALTREP’ vectors, and protection against long jumps and C++ exceptions.

Package	Version	Description
crayon	1.5.3	The crayon package provides functionality for colored terminal output using ANSI color codes, making it easy to create and combine colors and styles in R projects.
curl	6.4.0	This R package provides bindings to ‘libcurl’ for performing customizable HTTP/FTP requests, allowing responses to be processed in various ways, while also recommending the ‘httr2’ package for a more user-friendly experience.
datasetjson	0.3.0	This package facilitates the reading, constructing, and writing of CDISC Dataset JSON files while ensuring validation against the Dataset JSON schema.
desc	1.4.3	This package provides tools for reading, writing, creating, and manipulating DESCRIPTION files, facilitating the management of R packages.
diffdf	1.1.1	This package provides tools for comparing two data frames, detailing their differences and assisting in troubleshooting discrepancies.
digest	0.6.37	This package provides a function to create hash digests of various R objects using multiple algorithms, facilitating easy comparison of objects, along with capabilities for hash-based message authentication codes.

Package	Version	Description
dplyr	1.1.4	This package provides a fast and reliable tool for efficiently handling data frame-like objects, whether they are stored in memory or accessed externally.
emmeans	1.11.2	The package provides estimated marginal means (EMMs) for various statistical models, enabling users to compute contrasts, trends, and visualizations for better analysis of model outputs.
estimability	1.5.1	This package offers tools for assessing the estimability of linear functions of regression coefficients, including methods to handle non-estimable cases in regression analysis.
evaluate	1.0.4	This package provides tools for parsing and evaluating commands, enabling users to replicate R's command line behavior effortlessly.
fansi	1.0.6	This package provides counterparts to R's string manipulation functions that also consider ANSI text formatting control sequences, enabling accurate handling of styled text.
farver	2.1.2	The 'farver' package enables high-performance conversion and comparison of color representations across various color spaces, leveraging efficient C++ implementations for faster processing than R's standard 'convertColor' function.

Package	Version	Description
fastmap	1.2.0	The Fastmap package provides a memory-efficient key-value store implemented in C++ to avoid memory leakage associated with traditional R environments.
forcats	1.0.0	This package provides tools for reordering and modifying factor levels in R, allowing users to easily manipulate and organize factor data through various methods such as moving levels to the front, collapsing rare levels, and random shuffling.
formatters	0.5.11	This package offers a comprehensive framework for generating complex ASCII tables and includes formatters for converting values into display-ready strings.
fs	1.6.6	This package provides a cross-platform interface for file system operations, leveraging the ‘libuv’ C library for enhanced performance and compatibility.
generics	0.1.4	The generics package in R provides a collection of commonly used S3 generics to minimize package dependencies and conflicts.
ggplot2	3.5.2	ggplot2 is an R package that allows users to create complex and customizable visualizations by implementing a declarative system based on the principles of “The Grammar of Graphics.”

Package	Version	Description
ggsurvfit	1.1.0	This R package facilitates the creation of publication-ready survival endpoint figures using modular functions that integrate seamlessly with ‘ggplot2’ and its extensions.
glue	1.8.0	This R package implements interpreted string literals inspired by Python’s and Julia’s syntax, allowing for cleaner and more flexible string interpolation and documentation within the code.
gtable	0.3.6	The ‘gtable’ package in R facilitates the creation and manipulation of grid-based layouts for graphical objects (grobs), allowing for the easy construction of complex visual compositions.
haven	2.5.5	This package enables the import of various foreign statistical data formats into R by utilizing the embedded ‘ReadStat’ C library.
highr	0.11	This package offers syntax highlighting for R source code in LaTeX and HTML formats, with additional support for other languages through Andre Simon’s highlight package.
hms	1.1.3	This package implements an S3 class for effectively storing and formatting time-of-day values using the existing ‘difftime’ class in R.
htmltools	0.5.8.1	This package provides tools for generating and outputting HTML content effectively.

Package	Version	Description
huxtable	5.6.0	This package facilitates the creation of styled tables for data presentation, allowing for easy export to various formats and customizable features like borders, colors, and multi-row/column spanning.
isoband	0.2.7	This package provides a fast C++ implementation for generating contour lines (isolines) and contour polygons (isobands) from regularly spaced grid elevation data.
janitor	2.2.1	The janitor package in R provides user-friendly tools for cleaning and formatting data, generating frequency tables and crosstabs, and identifying duplicate records, all while integrating seamlessly with the tidyverse and the pipe operator.
jsonlite	2.0.0	The ‘jsonlite’ package in R provides efficient tools for parsing, generating, and manipulating JSON data, making it particularly valuable for statistical data analysis and web API interactions.
jsonvalidate	1.5.0	This package validates JSON data against specified JSON schemas using the ‘is-my-json-valid’ or ‘ajv’ libraries, supporting Drafts 04, 06, and 07 of the JSON schema.

Package	Version	Description
knitr	1.50	This package offers a versatile tool for generating dynamic reports in R through the use of Literate Programming techniques.
labeling	0.4.3	This package offers various algorithms for axis labeling to enhance data visualization in R.
lattice	0.22-7	The Lattice package in R provides a powerful framework for creating elegant, multivariate data visualizations inspired by Trellis graphics, catering to both typical and nonstandard graphical needs.
lifecycle	1.0.4	This package facilitates the management of exported functions' life cycles by providing standardized conventions, documentation badges, and clear deprecation warnings for users.
lubridate	1.9.4	The 'lubridate' package provides a user-friendly interface for parsing, manipulating, and extracting components from date-time objects in R, making it easy to work with temporal data.
magrittr	2.0.3	This package introduces a forward-pipe operator, <code>%>%</code> , allowing for streamlined chaining of commands by passing values or expressions to subsequent function calls.

Package	Version	Description
MASS	7.3-65	This package provides functions and datasets to support the methodologies presented in Venables and Ripley's "Modern Applied Statistics with S," facilitating statistical analysis and modeling.
Matrix	1.7-3	This package provides a comprehensive collection of matrix classes, both sparse and dense, encompassing various types such as general, symmetric, triangular, and diagonal matrices, and offers efficient operations leveraging high-performance libraries like BLAS, LAPACK, and SuiteSparse.
memoise	2.0.1	This package allows for caching the results of a function so that subsequent calls with the same arguments return the previously computed value, improving efficiency.
metacore	0.2.0	This package creates an immutable container for holding metadata to enhance programming activities and improve the functionality of other packages within the clinical programming workflow.
metatools	0.1.6	This package utilizes metadata from 'metacore' objects to verify and construct associated metadata columns.

Package	Version	Description
mgcv	1.9-3	This package provides tools for fitting generalized additive (mixed) models, including multiple smoothing parameter estimation and support for various distributions, enhancing Bayesian inference and model flexibility.
mvtnorm	1.3-3	This package provides functions for computing multivariate normal and t probabilities, quantiles, random deviates, and densities, along with log-likelihoods and score functions for multivariate Gaussian models and Gaussian copulae, all through Cholesky factorization of covariance or precision matrices for various types of data.
nlme	3.1-168	This package facilitates the fitting and comparison of Gaussian linear and nonlinear mixed-effects models for enhanced statistical analysis.
numDeriv	2016.8-1.1	This R package offers methods for accurately calculating first and second order numerical derivatives using techniques like Richardson's extrapolation and complex step differentiation, while also providing faster but less accurate simple difference methods for cross-checking.

Package	Version	Description
patchwork	1.3.1	The ‘patchwork’ package enhances ‘ggplot2’ by enabling the composition of multiple plots using mathematical operators, allowing for complex plot arrangements with ease.
pharmaRTF	0.1.4	This R package provides an enhanced RTF wrapper that enables the addition of metadata and formatting features to tables generated by existing R table packages, facilitating their use in regulatory reports.
pillar	1.11.0	This package offers generics for formatting data columns with a wide range of colors, enhancing the visual presentation in modern terminals.
pkgbuild	1.4.8	This package provides essential functions for building R packages, including locating necessary compilers and configuring the PATH for optimal usage in various environments.
pkgconfig	2.0.3	This package allows users to set configuration options specifically for individual R packages, ensuring that changes apply only to the designated package without affecting others.
pkgload	1.4.0	The package simulates the installation and attachment process of R packages, facilitating rapid iteration during package development.

Package	Version	Description
prettyunits	1.2.0	This package provides human-readable formatting for various quantities, including time intervals, byte sizes, significant digits, p-values, colors, and numerical values, transforming them into more accessible representations.
processx	3.8.6	The processx package in R enables the management of system processes in the background, allowing users to check their status, wait for completion, retrieve exit statuses, kill processes, and handle standard output and error through non-blocking connections and polling.
progress	1.2.3	This package offers customizable progress bars that show percentage, elapsed time, and estimated completion time, compatible with various R environments including terminals and IDEs like RStudio and Emacs.
ps	1.9.1	This package allows users to list, query, and manipulate all system processes across Windows, Linux, and macOS platforms.
purrr	1.1.0	This package provides a comprehensive and coherent toolkit for functional programming in R, enabling efficient and expressive coding practices.

Package	Version	Description
r2rtf	1.1.4	This package allows users to create production-ready Rich Text Format (RTF) tables and figures with customizable formatting options.
R6	2.6.1	The package allows for the creation of lightweight reference classes in R, featuring public and private members, inheritance, and no dependency on S4 classes.
RColorBrewer	1.1-3	This R package offers a collection of color schemes specifically designed for maps and other graphics, based on the work of Cynthia Brewer.
Rcpp	1.1.0	The ‘Rcpp’ package enables seamless integration between R and C++ by providing R functions and C++ classes that facilitate mapping R data types to C++ equivalents, enhancing code development and integration with third-party libraries.
readr	2.1.5	The ‘readr’ package offers a fast and user-friendly solution for reading rectangular data formats such as CSV, TSV, and FWF, with flexible parsing capabilities and robust error handling.
readxl	1.4.5	This R package imports Excel files, supporting both ‘xls’ and ‘xlsx’ formats across multiple operating systems without external dependencies.

Package	Version	Description
rematch	2.0.0	This package provides a simplified interface for extracting matches and captured groups from regular expressions applied to character vectors, serving as a convenient wrapper around the <code>regexr</code> function in R.
renv	1.1.4	renv is an R package that facilitates dependency management by allowing users to create project-specific libraries, save their state to a lockfile, and restore them as needed for enhanced isolation, portability, and reproducibility.
rlang	1.1.6	This package provides a comprehensive toolbox for working with base R types and features, core condition handling, and key ‘Tidyverse’ functionalities, particularly focusing on tidy evaluation.
roxygen2	7.3.2	The package facilitates the generation of R package documentation, ‘NAMESPACE’ files, and collation fields using specially formatted comments, allowing for seamless integration of documentation with code updates.
rprojroot	2.1.0	This package provides robust, reliable, and flexible methods for constructing file paths relative to a defined project root directory, which is identified based on the presence of specific files.

Package	Version	Description
rtables	0.6.13	The ‘rtables’ package provides a framework for creating complex, hierarchical reporting tables that support multi-level tabulations, contextual summaries, and flexible layout designs with a convenient pipe-able interface.
scales	1.4.0	This package facilitates the mapping of data to graphical aesthetics by offering automated methods for determining breaks and labels for axes and legends in visualizations.
snakecase	0.11.1	This package provides a convenient and adaptable tool for parsing and converting strings into various casing styles, such as snake_case and camelCase.
stringi	1.8.7	The stringi package provides a comprehensive suite of tools for efficient string manipulation and processing, including pattern searching, random string generation, Unicode handling, and date-time formatting across various locales and platforms.
stringr	1.5.1	This R package provides a consistent and user-friendly interface for the ‘stringi’ package, ensuring uniform handling of NA values and zero-length vectors across its functions.

Package	Version	Description
survival	3.8-3	This package provides essential tools for survival analysis, including the creation of survival objects, Kaplan-Meier and Aalen-Johansen curve estimation, Cox proportional hazards models, and parametric accelerated failure time models.
tibble	3.3.0	This R package offers the ‘tibble’ class, which enhances data frames with stricter validation and improved formatting for a more user-friendly experience.
tidyr	1.3.1	The ‘tidyr’ package provides tools for transforming data into a tidy format, facilitating tasks such as reshaping, nesting, and extracting values, while also addressing missing data.
tidyselect	1.2.1	This package provides a backend for implementing select-like functions in R, ensuring compatibility with the selection interfaces of the ‘tidyverse’ ecosystem.
timechange	0.3.0	This R package provides efficient tools for manipulating date-time objects, offering features like time-zone adjustments, daylight saving time considerations, and various operations for updating and rounding date-time components.

Package	Version	Description
Tplyr	1.2.1	This package simplifies data manipulation for generating clinical summaries with a focus on traceability.
tzdb	0.5.0	This R package offers an updated interface to the IANA Time Zone Database and the ‘date’ library, facilitating comprehensive date and time manipulations, including support for time zone adjustments and calendar-specific calculations.
utf8	1.2.6	This package processes and prints UTF-8 encoded international text, facilitating input, validation, normalization, encoding, formatting, and display of Unicode characters.
V8	6.0.4	This R package provides an interface to the V8 JavaScript engine, enabling users to execute JavaScript and WebAssembly code seamlessly within R.
vctrs	0.6.5	This package provides tools for consistent type coercion and size recycling, integrating concepts of type and size stability to enhance the analysis of function interfaces in R.

Package	Version	Description
viridisLite	0.4.2	The ‘viridis’ package provides color maps that enhance graph readability for individuals with color vision deficiencies and ensure perceptual uniformity for both color and black-and-white printing, along with integrated support for ‘ggplot2’ visualizations.
vroom	1.6.5	The ‘vroom’ package in R efficiently reads and writes data files like CSV and TSV by using quick indexing and lazy loading, allowing only the necessary data to be accessed, while also enabling parallel writing and asynchronous disk operations.
withr	3.0.2	This package offers a collection of functions that allow users to safely and temporarily modify the global state while executing code, streamlining development tasks without extensive dependencies.
xfun	0.52	This package provides various miscellaneous functions that are commonly utilized in other packages maintained by Yihui Xie.
xml2	1.3.8	This package provides an interface for working with XML data through XPath expressions, supports XML schema validation, and facilitates XSLT transformations via an additional package.

Package	Version	Description
xportr	0.4.3	This package provides tools for creating CDISC-compliant data sets and verifying their compliance with CDISC standards.
yaml	2.3.10	This package provides an R interface to the ‘libyaml’ library, enabling parsing and emitting of YAML 1.1 data formats.
yyjsonr	0.1.21	This R package provides a fast parser, generator, and validator for converting JSON, NDJSON, and GeoJSON data to and from R objects, supporting various data types and structures.

9 Appendix

(insert text here or remove this section)

Legacy Data Conversion Plan and Report Appendix

10 Purpose

The purpose of this appendix is to document the traceability of key output analysis results with ADaM when the analysis results were generated using a legacy process.

Because of transformations required during ADaM conversion, some of the terms, categories and data formats used in the tabulation data have been translated into CDISC standard formats in the ADaM data. This appendix identifies differences between the legacy analysis and ADaM data, and explains how ADaM represents the equivalent data.

11 Conversion Data Flow

The legacy data was converted to SDTM/ADaMas described in the following data flow diagram.

Rationale:

(Text here)

12 Converted Data Summary

12.1 Issues Encountered and Resolved

- (Text and/or table here)
- (Text and/or table here) # Traceability Data Flow

The legacy data traceability from collection to submission is described in the following data flow diagram.

13 Outstanding Issues

- (Text and/or table here)
- (Text and/or table here)